A Data sources and cohorts

MMRF CoMMpass, interim analysis 24 (IA24)

MMRF CoMMpass (IA24)
1,143 subject records

Clinical data
diagnosis · laboratory · treatment · response · adverse events · symptoms

Bulk RNA sequencing
TPM → ssGSEA (MSigDB Hallmark): 50 pathway scores per RNA sample

Visit pairs V(t) → V(t+1)
13,451 pairs
1,025 patients with ≥1 pair

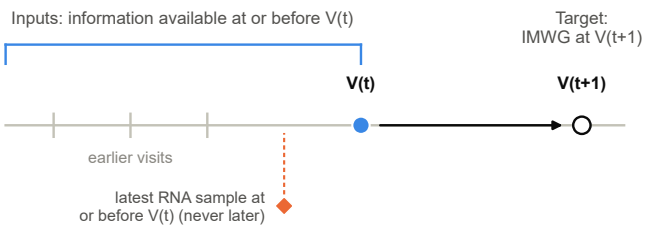
RNA available at V(t)
9,173 pairs · 707 patients
(68% of pairs)

Six-class response task
13,451 pairs · 1,025 patients
IMWG: PD/SD/PR/VGPR/CR/sCR

Binary improvement task
12,298 pairs · 1,018 patients
excludes 1,153 pairs with sCR at V(t)

Survival cohort: 674 patients
RNA by day 0 · PFS 459 events · OS 247 deaths

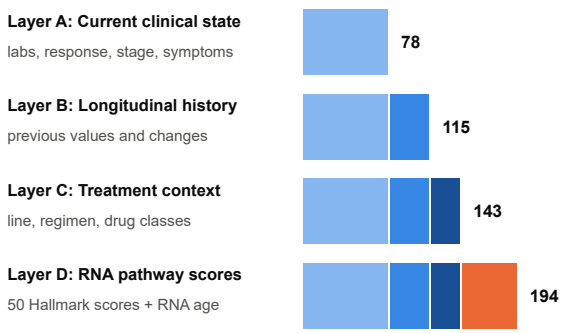
B Visit-pair construction



Six-class: PD, SD, PR, VGPR, CR or sCR at V(t+1).
Binary: V(t+1) is a better IMWG category than V(t).
Every pair of a patient stays in one data partition.

C Information layers A–D

Nested feature sets, not algorithms

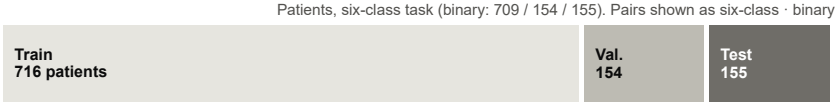


Bars: cumulative source input features, coloured by the layer that adds them
(D adds 50 pathway scores + RNA age, plus a derived RNA-availability indicator).

D Patient-level partitions and modelling framework

Assigned once; shared by every layer and algorithm; no patient is in more than one partition

Patients, six-class task (binary: 709 / 154 / 155). Pairs shown as six-class · binary



Train: tuning and fitting
Preprocessing fitted; hyperparameters chosen by 5-fold patient-grouped CV on these patients only
Pairs: six-class 9,539 · binary 8,709

Validation
Descriptive only: no tuning, threshold or calibration
Pairs: 1,833 · 1,659

Test
Scored once; patient-level bootstrap CIs
Pairs: 2,079 · 1,930

Modelling framework

- Logistic Regression
- LightGBM
- XGBoost

4 layers × 3 algorithms × 2 tasks = 24 models

Binary: AUPRC (test improvement rate 18.3%)
Six-class: macro-F1 (argmax)
Also AUROC, Brier score, RPS, log loss

E Evaluation, downstream analyses and validation status

All evaluations are internal to CoMMpass; no external validation cohort was used

Analysis	Patients and design	Validation status
Main prediction (24 models), calibration and SHAP	Test: 155 patients, scored once. Tuning on train patients only	Internal held-out test split; calibration and SHAP descriptive
RNA contribution, uncertainty and subgroup analyses	Development: 716 train patients (CV reference). Evaluation: 213 validation + test patients with RNA; locked models, scored once	Post hoc, exploratory; earlier pooled results were seen first; not independent
Train-selected pathway composite (association)	Selection and discovery: 371 train patients. Evaluation: 160 validation + test patients with RNA (76 + 84)	Post hoc, exploratory; earlier pooled results were seen first; not independent
Survival (PFS, OS) and pathway coefficients	Development: 473 train patients (CV; Fig. 6). Evaluation: 201 validation + test patients (101 + 100); frozen models, scored once	Post hoc, exploratory; earlier pooled results were seen first; not independent
Geographic hold-out (US → Spain, Canada)	Tuning and fit: 439 US train-partition patients; threshold, calibration: 96 such patients. Scored once: 116 Spain/Canada	Post hoc, exploratory; 236 US validation/test-partition patients and Italy excluded

Main models and every redesigned analysis (RNA contribution, pathway composite, survival, geographic): tuning, selection, fitting and preprocessing use train-partition patients only; held-out patients are scored once. No decision threshold or recalibration enters any reported metric of the main models (probabilities are shown raw; validation patients are descriptive only); the geographic model's threshold and calibration use train-partition patients. The RNA-contribution, pathway, survival and geographic analyses are post hoc and exploratory (earlier pooled results were seen first); nothing here is externally validated, and CIs are descriptive and unadjusted.